

SAT 7

Math Module 2

Question # ID
SAT7 M 2.1 a07ma201

A total of 165 people contributed to a charity event as either a donor or a volunteer. 130 people contributed as a donor. How many people contributed as a volunteer?

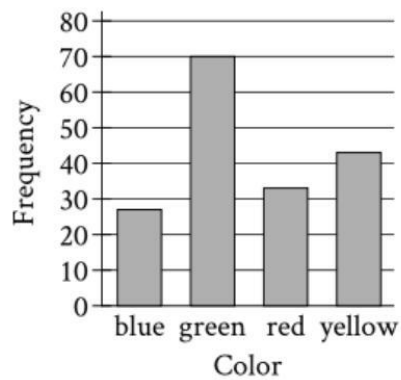
(A) 35

(B) 130

(C) 165

(D) 330

SAT7 M 2.2 a07ma202



A data set consists of 173 colors. The bar graph shows the number of times each color appears in the data set. Which color appears 70 times?

(A) Blue

(B) Green

(C) Red

(D) Yellow

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Question # ID
SAT7 M 2.3 a07ma203

What value of p satisfies the equation $2p + 275 = 325$?

(A) 5

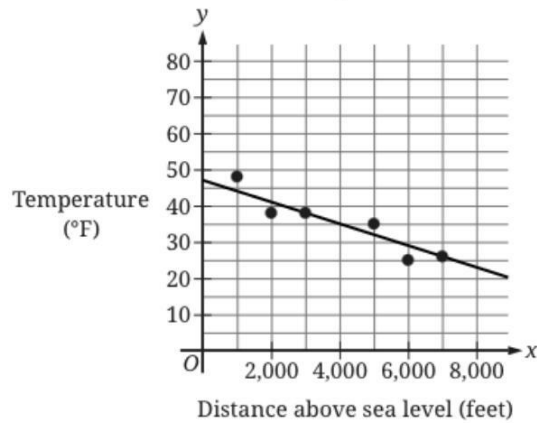
(B) 25

(C) 48

(D) 300

SAT7 M 2.4 a07ma204

The scatterplot shows the temperature, in degrees Fahrenheit ($^{\circ}\text{F}$), and the distance above sea level, in feet, measured at 6 locations on Mount Jefferson. A line of best fit is also shown.



At a distance of 4,000 feet above sea level, what is the temperature, in $^{\circ}\text{F}$, predicted by the line of best fit?

(A) 47

(B) 35

(C) 25

(D) 0

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Question # ID

SAT7 M 2.5 a07ma205

Rectangle P has an area of **72** square inches. If a rectangle with an area of **20** square inches is removed from rectangle P, what is the area, in square inches, of the resulting figure?

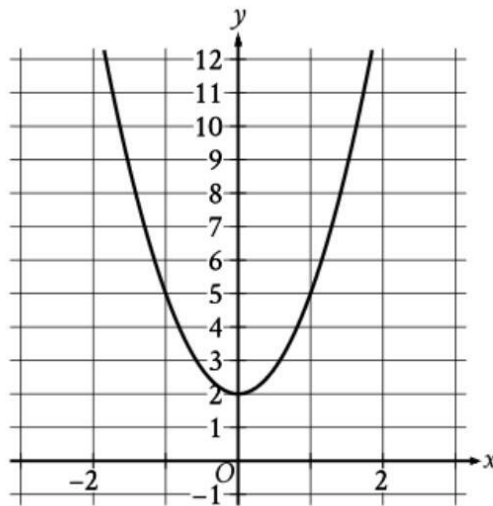
(A) 92

(B) 84

(C) 80

(D) 52

SAT7 M 2.6 a07ma206



The graph of the quadratic function $y = f(x)$ is shown. What is the vertex of the graph?

(A) $(0, -2)$

(B) $(0, -3)$

(C) $(0, 2)$

(D) $(0, 3)$

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SAT7 M 2.7 a07ma207

The number of raccoons in a 131-square-mile area is estimated to be 2,358. What is the estimated population density, in raccoons per square mile, of this area?

☐ (A) 18☐ (B) 131☐ (C) 149☐ (D) 2,376

SAT7 M 2.8 a07ma208

If $8x = 6$, what is the value of $72x$?

☐ (A) 3☐ (B) 15☐ (C) 54☐ (D) 57

SAT7 M 2.9 a07ma209

 $-11, -9, 26$

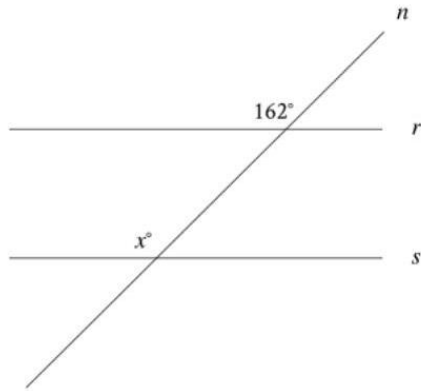
A data set of three numbers is shown. If a number from this data set is selected at random, what is the probability of selecting a positive number?

☐ (A) 0☐ (B) $\frac{1}{3}$ ☐ (C) $\frac{2}{3}$ ☐ (D) 1

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SAT7 M 2.10 a07ma210



Note: Figure not drawn to scale.

In the figure, line n intersects lines r and s . Line r is parallel to line s . What is the value of x ?

SAT7 M 2.11 a07ma211

Which expression is equivalent to $23x^3 + 2x^2 + 9x$?

(A) $23x(x^2 + 2x + 9)$

(B) $9x(23x^3 + 2x^2 + 1)$

(C) $x(23x^2 + 2x + 9)$

(D) $34(x^3 + x^2 + x)$

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Question # ID
SAT7 M 2.12 a07ma212

Which expression is equivalent to $(9x^3 + 5x + 7) + (6x^3 + 5x^2 - 5)$?

(A) $15x^6 + 5x^2 - 5x - 35$

(B) $15x^3 + 10x^2 + 2$

(C) $15x^6 + 5x^2 + 5x + 2$

(D) $15x^3 + 5x^2 + 5x + 2$

SAT7 M 2.13 a07ma213

$$f(x) = 45x + 600$$

The function f gives the monthly fee $f(x)$, in dollars, a facility charges to keep x crates in storage. What is the monthly fee, in dollars, the facility charges to keep 50 crates in storage?

SAT7 M 2.14 a07ma214

There are 450 tiles in a box. Of these tiles, 6% are black. How many black tiles are in the box?

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Question # ID
SAT7 M 2.15 a07ma215

An investment account was opened with an initial value of \$890. The value of the account doubled every 10 years. Which equation represents the value of the account $M(t)$, in dollars, t years after the account was opened?

(A) $M(t) = 890\left(\frac{1}{2}\right)^{\frac{t}{10}}$

(B) $M(t) = 890\left(\frac{1}{10}\right)^{\frac{t}{2}}$

(C) $M(t) = 890(2)^{\frac{t}{10}}$

(D) $M(t) = 890(10)^{\frac{t}{2}}$

SAT7 M 2.16 a07ma216

$$y < x$$

$$x < 22$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

(A)

x	y
19	18
20	19
21	20

(C)

x	y
23	22
24	23
25	24

(B)

x	y
19	20
20	21
21	22

(D)

x	y
23	24
24	25
25	26

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Question # ID

SAT7 M 2.17 a07ma217

A line in the xy -plane has a slope of 9 and passes through the point $(0, -5)$. The equation $y = px + r$ defines the line, where p and r are constants. What is the value of p ?

SAT7 M 2.18 a07ma218

The function f is defined by $f(x) = \frac{1}{2}(x + 6)$. What is the value of $f(4)$?

☐ (A) 20☐ (B) 12☐ (C) 10☐ (D) 5

SAT7 M 2.19 a07ma219

What is the slope of the graph of $10x - 5y = -12$ in the xy -plane?

☐ (A) -2 ☐ (B) $-\frac{5}{6}$ ☐ (C) $\frac{5}{6}$ ☐ (D) 2

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Question # ID

SAT7 M 2.20 a07ma220

$$y = 6x + 3$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. Which equation could be the second equation in this system?

(A) $y = 2(6x) + 3$

(B) $y = 2(6x + 3)$

(C) $2(y) = 2(6x) + 3$

(D) $2(y) = 2(6x + 3)$

SAT7 M 2.21 a07ma221

The function f is defined by $f(x) = |x - 4x|$. What value of a satisfies $f(5) - f(a) = -15$?

(A) -20

(B) 5

(C) 10

(D) 45

SAT7 M 2.22 a07ma222

In August, a car dealer completed 15 more than 3 times the number of sales the car dealer completed in September. In August and September, the car dealer completed 363 sales. How many sales did the car dealer complete in September?